



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE (24AD3107)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	FinSight: Explainable Credit Risk Analytics and Loan Decision Support System	
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Abstract

Credit risk assessment helps financial institutions evaluate loan applicants and manage the risk of repayment failure. Incorrect classifications can result in financial losses or the rejection of creditworthy applicants. Predictive systems also need to provide understandable explanations so that reviewers can assess their outputs. This project proposes FinSight, an explainable credit risk analytics and loan decision support system that combines data preparation, machine learning, and interactive visualisation.

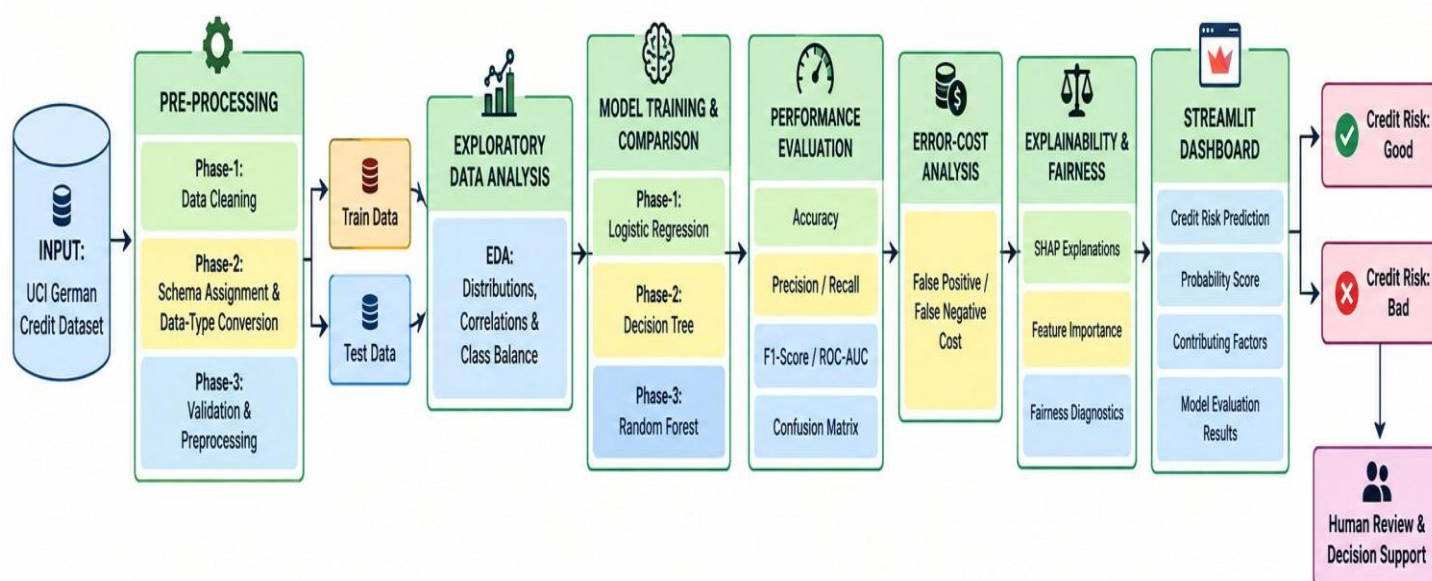
The project uses the UCI Statlog German Credit dataset, containing 1,000 applicant records with 20 predictor attributes and a credit risk label. These attributes describe information such as credit history, loan duration, credit amount, savings, employment, and housing. A reproducible Python pipeline will acquire and store the original dataset locally. Pandas will parse the whitespace-separated records, assign meaningful column names, and convert numerical and categorical fields into appropriate data types. Validation will check missing values, duplicate records, column structure, target labels, and categorical codes. Automated data-quality reports and a data dictionary will document the dataset.

Exploratory data analysis will examine feature distributions, relationships, and class imbalance. The dataset will be divided into training and testing sets, with preprocessing fitted only on the training data to prevent data leakage. Categorical encoding and appropriate numerical scaling will prepare the features for modelling. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting classifiers will be trained and compared using precision, recall, F1-score, ROC-AUC, and confusion matrices. Evaluation will also consider the unequal costs of classification errors, particularly the consequences of classifying a high-risk applicant as low risk.

SHAP will be used to explain the factors influencing overall model behaviour and individual predictions. Fairness diagnostics will examine potential differences in model performance across relevant applicant groups, taking account of limitations in the dataset’s demographic coding. A Streamlit dashboard will display applicant inputs, predicted credit risk, estimated probabilities, contributing factors, and analytical visualisations.

The expected outcome is a reproducible application that supports transparent credit risk analysis and demonstrates the complete data science workflow from raw-data preparation to model interpretation. FinSight will provide advisory outputs for human review, with the final lending decision remaining under human supervision.

Project Architecture/Project Overview:



Signature of the Guide

HOD

